$\begin{cases} Ax = b \end{cases} \rightarrow$ Solvable or $\exists v$ with $v'A = 0$, $v'b \neq 0$.

$\begin{cases} Ax = b \\ x \geq 0 \end{cases} \rightarrow$ Solvable or $\exists v$ with $v'A \geq 0$, $v'b < 0$

$\begin{cases} Ax \leq b \\ x \geq 0 \end{cases} \rightarrow$ Solvable, or $\exists v$ with $v'A \geq 0$, $v'b < 0$

$Ax \leq b$, $z = b - Ax$

$Ax + z = b$, $z \geq 0$

$\tilde{x} = \begin{bmatrix} z \\ x \end{bmatrix}$ $\quad \tilde{A} = [I \; A]$

$\tilde{A}\tilde{x} = [I \; A] \begin{bmatrix} z \\ x \end{bmatrix} = Iz + Ax = b$

$v'\tilde{A} = v'[I \; A] = [v' \; v'A] \geq 0$

$v'b < 0$

$A = \begin{bmatrix} & \\ & \end{bmatrix}$

$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, $c = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$

$y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$ $\quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$

' is transpose

$s = \inf_{x \in \mathbb{R}^n} c'x$ minimum dot product with c

$S = \sup_{y \in \mathbb{R}^m} y'b$ maximum dot product with b

$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \cdot \begin{bmatrix} E \\ F \end{bmatrix} = \begin{bmatrix} AE + BF \\ CE + DF \end{bmatrix}$

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Theorem 3.2. *(Duality). Suppose there exist feasible vectors x and y. Then S and s are finite and $S = s$.*

Proof. Since x must be a vector with non negative components then by multiplying both sides the first inequality constraint for the sup problem we get

$$c'x \geq y'Ax \geq y'b \tag{3.3}$$

or equivalently $s \geq S$. To prove the equality in between s and S it is sufficient to show the existence of a feasible y for the sup problem in (3.1) such that

$$y'b \geq s. \tag{3.4}$$

In order to consider a feasible y for the sup problem that at the same time satisfies eq. (3.4) we will combine the inequalities $A'y \leq c$ (equivalent to $y'A \leq c$) and $y \geq 0$ with (3.4) as follow

$$\begin{bmatrix} A' \\ -b \end{bmatrix} y \leq \begin{bmatrix} b \\ -s \end{bmatrix}, \qquad y \geq 0 \tag{3.5}$$

Let's show the existence of such y by contradiction. Let's suppose that there is a feasible y for the sup problem that does not satifies (3.4). Then such y does not satisfies (3.5) and according to theorem (2.5) there exist a vector v and a scalar a such that

$$[v' \; a] \geq 0, \qquad [v' \; a] \begin{bmatrix} A' \\ -b \end{bmatrix} \geq 0, \qquad [v' \; a] \begin{bmatrix} b \\ -s \end{bmatrix} < 0. \tag{3.6}$$

Clearly if $a = 0$ the system (3.6) reduces to

$$v \geq 0, \qquad v'A' \geq 0, \qquad v'b < 0 \tag{3.7}$$

meaning that y (again in virtue of theorem (2.5)) is not feasible contrary to our assumption. This means that a has to be positive rather than non negative. Without lost of generality let $a = 1$. With this choice eq. (3.6) leads to the contradiction

$$v \geq 0, \qquad v'A \geq b, \qquad v'b < s. \tag{3.8}$$

saying that exists a feasible vector v for the infimum problem satisfying $v'b < \inf_x c'x$. This contradiction is consequence of the fact that we supposed the existence of a feasible y not satifying (3.4). Summarizing, we can find a feasible y satisfying (3.4) and $S \geq s$. This fact, together with (3.3), proves that $S = s$. □

Remark 3.9. The minimization and maximization problem in (3.3) are often called *primal* and *dual* problem respectively.

Remark 3.10. Clearly s is finite since the entries of b, A and c are finite. The inequality (3.4) is indeed an equality since otherwise would contradict the fact that $S = s$. This says that the supremum in (3.3) is achieved, namely that the supremum is indeed a maximum. Anagolously one can show that the infimum is indeed a minimum by turning the supremum into an infimum (this can be realized replacing A, c and b with $-A$, $-c$ and $-b$).

Remark 3.11. By using theorem (2.2) rather than its generalization (2.5) we can, repeating the same proof for the theorem (3.2), show that given the primal and

Theorem 2.5. *Either the problem*

$$\begin{cases} Ax \leq b \\ x \geq 0 \end{cases} \tag{2.6}$$

has a solution or there exists a vector $v \geq 0$ such that $v'A \geq 0$ and $v'b < 0$.

Proof. It is sufficient to notice that the problem (2.5) can be recasted in the form of (2.2) by using a slack variable z. By defining

$$z = b - Ax, \qquad \tilde{x} = \begin{bmatrix} z \\ x \end{bmatrix} \quad \text{and} \quad \tilde{A} = [I \;\; A] \tag{2.7}$$

we can rewrite (2.3) as

$$\begin{cases} \tilde{A}\tilde{x} = b \\ \tilde{x} \geq 0 \end{cases} \tag{2.8}$$

The equations in (2.8) have the same form of those one in (2.3) so we can use the theorem (2.2) to assert that either (2.8) has a solution or there exists v s.t. $v'\tilde{A} \geq 0$ and $v'b < 0$. This is equivalent to the statement of the theorem. □

$\begin{cases} Ax \geq b \\ x \geq 0 \\ s = \inf_x c'x \end{cases}$ $\qquad \begin{cases} y'A \leq c \\ y \geq 0 \\ S = \sup_y y'b \end{cases}$

If $\begin{cases} Ax \geq b \\ x \geq 0 \end{cases}$ is solvable, then s as the smallest dot product of the "feasible" solutions.

Ax big, minimize $c'x$

Find the "lowest cost" x which is sufficient $(Ax \geq b)$

$y'A$ small, maximize $y'b$

Find the "highest output" y which is on-budget $(y'A \leq c)$

Theorem 3.2. *(Duality). Suppose there exist feasible vectors x and y. Then S and s are finite and $S = s$.*

I.e., If both halves are feasible, then their results are equal (and finite)!

Proof. Since x must be a vector with non negative components then by multiplying both sides the first inequality constraint for the sup problem we get

$$c'x \geq y'Ax \geq y'b \tag{3.3}$$

or equivalently $s \geq S$. To prove the equality in between s and S it is sufficient to show the existence of a feasible y for the sup problem in (3.1) such that

$$y'b \geq s. \tag{3.4}$$

In order to consider a feasible y for the sup problem that at the same time satisfies eq. (3.4) we will combine the inequalities $A'y \leq c$ (equivalent to $y'A \leq c$) and $y \geq 0$ with (3.4) as follow

$$\begin{bmatrix} A' \\ -b \end{bmatrix} y \leq \begin{bmatrix} b \\ -s \end{bmatrix}, \qquad y \geq 0 \tag{3.5}$$

Let's show the existence of such y by contradiction. Let's suppose that there is a feasible y for the sup problem that does not satifies (3.4). Then such y does not satisfies (3.5) and according to theorem (2.5) there exist a vector v and a scalar a such that

$$[v' \; a] \geq 0, \qquad [v' \; a] \begin{bmatrix} A' \\ -b \end{bmatrix} \geq 0, \qquad [v' \; a] \begin{bmatrix} b \\ -s \end{bmatrix} < 0. \tag{3.6}$$

Clearly if $a = 0$ the system (3.6) reduces to

$$v \geq 0, \qquad v'A' \geq 0, \qquad v'b < 0 \tag{3.7}$$

meaning that y (again in virtue of theorem (2.5)) is not feasible contrary to our assumption. This means that a has to be positive rather than non negative. Without lost of generality let $a = 1$. With this choice eq. (3.6) leads to the contradiction

$$v \geq 0, \qquad v'A \geq b, \qquad v'b < s. \tag{3.8}$$

$\exists x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$,

$x, y \geq 0$ $\quad Ax \geq b$ $\quad y'A \leq c'$

If $x \geq 0$, $u \leq v$,

$u'x \leq v'x$

$x'u \leq x'v$

x is feasible, y is feasible

$Ax \geq b$, $y'A \leq c'$

$y'Ax \geq y'b$, $y'Ax \leq c'x$

since $y \geq 0$, since $x \geq 0$

So $c'x \geq y'Ax \geq y'b$

So $y'Ax$ is a lower bound on $\inf_x c'x$

$y'Ax$ is an upper bound on $\sup_y y'b$

So $\inf_x c'x \geq \sup_y y'b$

$\forall x_0, y_0$ $\quad c'x_0 \geq \inf_x c'x \geq y_0'Ax_0 \geq \sup_y y'b \geq y_0'b$

So S, s finite and $s \geq S$

We know $s \geq y'b$ $\forall y$

NTS $\exists y_0$ s.t. $s \leq y_0'b$

$\Rightarrow s = y_0'b \leq S$, $s \geq S$

$\Rightarrow s = S$.

$\begin{cases} y \geq 0 \\ y'A \leq c \\ y'b \geq S \end{cases}$ If feasible, then there is a feasible y which exceeds S.